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CAN ESG MITIGATE CREDIT RISK?



Evidence from Emerging Market Banks: A System GMM Approach



Dataset Selection Motivation

Research Relevance

Directly aligned with the research objective

1. Contains ESG Score (ESGS) and Credit Risk (NPL)
2. Includes both bank-specific and macroeconomic variables
3. Suitable for examining the ESG–Credit Risk relationship

Suitable for System GMM

Large N, Small T Structure

- 156 Banks
 - 7 Years
 - 1092 Observations
- ⇒ Ideal panel structure for Dynamic Panel and System GMM estimation.

Dynamic Nature of Credit Risk

- Current NPL depends on past NPL
 - Credit risk exhibits persistence over time
- ⇒ Dynamic model is more appropriate than static models.

Endogeneity Issue

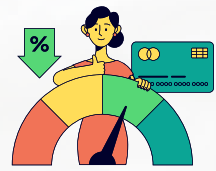
- ESG may affect NPL
 - NPL may also affect ESG
 - Potential omitted variables
- ⇒ System GMM helps address endogeneity and improve estimation reliability.

This dataset provides an ideal setting to examine whether ESG performance can help reduce credit risk in emerging market banks.



Research Variables

1 Dependent Variable



Credit Risk

- Measured by NPL Ratio
- Proxy for bank credit risk

2 Main Independent Variable



ESG Performance

- ESG Score (ESGS)
- Environmental
- Social
- Governance

3 Control Variables



Bank-specific Factors

- ETA
- CTI
- TDTA
- REVDIV
- LOANGR
- LNTA



Macroeconomic Factors

- EG
- INF
- HHII





RESEARCH QUESTION

To what extent does ESG performance mitigate credit risk in emerging market banks?

- ESG performance is expected to reduce credit risk in emerging market banks. Higher ESG scores are associated with better governance, stronger risk management, and lower non-performing loans, leading to a reduction in overall credit risk.

**Challenge
Solution**



System GMM



Credit Risk



Exploratory Data Analysis

DATA SOURCE & PRE-PROCESSING

- Balanced panel of 156 banks over 7 years (2016–2022): $N * T = 1092$ observations.
- Bank_ID and Country_ID encoded numerically for panel alignment and processing.
- Widespread missing data in ESGS/ESGcom/ESGC (143 points) and NPL (19 points).
- Resolved via Multiple Imputation by MICE → 0 missing values retained.
- Cleaned matrix fully preserves multivariate distributions, securing a robust basis for modeling.

```
Number of banks (N): 156
Number of years (T): 7
Expected observations (N x T): 1092
Actual observations: 1092
```

```
=== TÌNH TRẠNG DỮ LIỆU BAN ĐẦU (TRƯỚC KHI ĐIỀN KHUYẾT MICE) ===
Số lượng ô trống chi tiết ở các cột ban đầu bị thiếu:
```

	0
DTD_PB	78
DTC_PB	78
ESGS	143
ESGcom	143
ESGC	143
E	143
S	143
G	143
ETA	5
NPL	19
CTI	5
TDTA	5
REVDIV	5

```
=== KIỂM TRA TÌNH TRẠNG DỮ LIỆU SAU KHI ĐIỀN KHUYẾT (MICE) ===
Tổng số ô trống còn sót lại trên toàn bảng df_imputed: 0
```

```
Số lượng ô trống chi tiết ở các cột ban đầu bị thiếu:
```

	0
DTD_PB	0
DTC_PB	0
ESGS	0
ESGcom	0
ESGC	0
E	0
S	0
G	0
ETA	0
NPL	0
CTI	0
TDTA	0

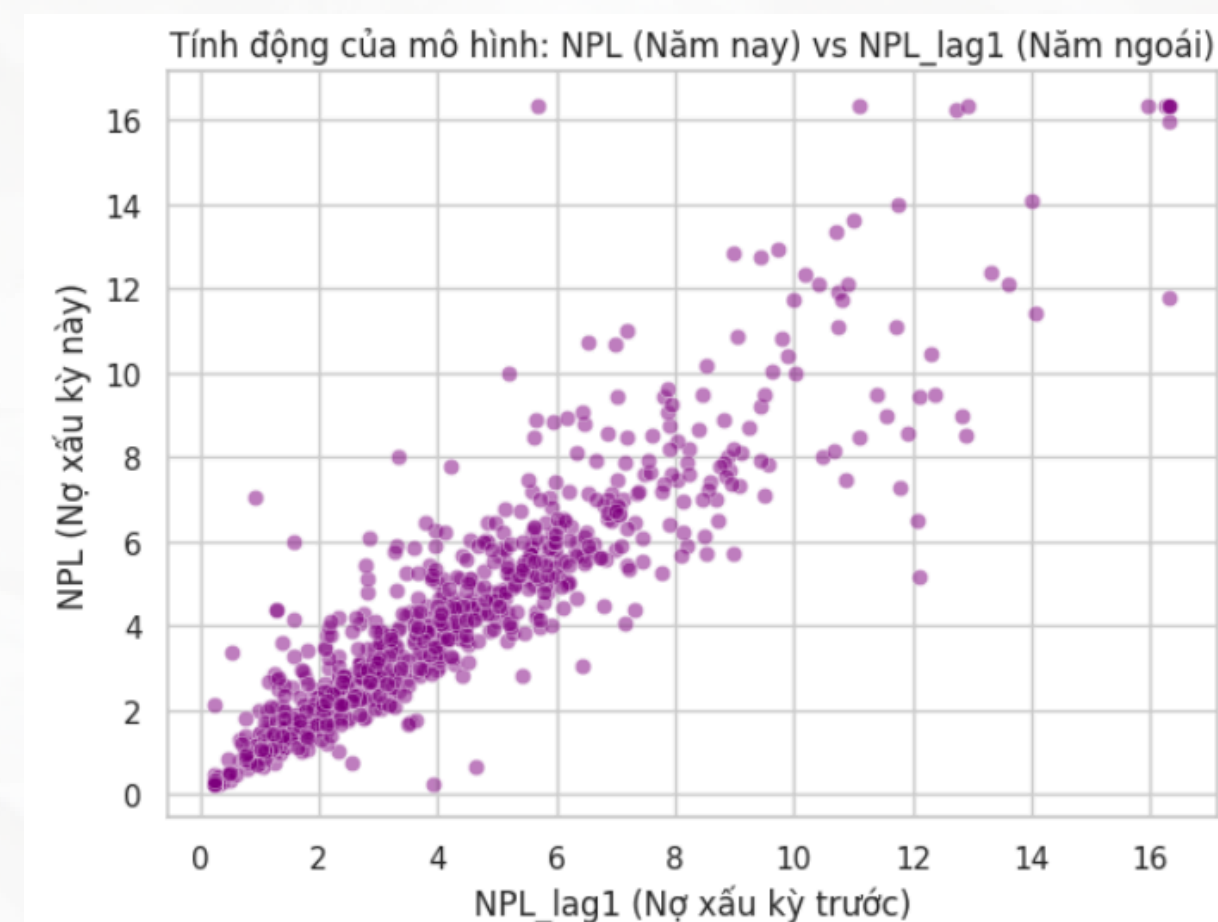
Exploratory Data Analysis

PRE-MODELING DIAGNOSTICS & JUSTIFICATION

- Zero-lag ADF tests confirm 139 banks (89.10%) exhibit non-stationary NPL series.
- NPL_{i,t} vs. NPL_{i,t-1} plot shows strong positive linear clustering -> credit risk is path-dependent.
- Variance Inflation Factors are optimal; all VIF < 5.0.
- Persistent, non-stationary NPL combined with dynamic lag induces severe endogeneity.
- Static frameworks (OLS, FE, RE) are inherently biased.

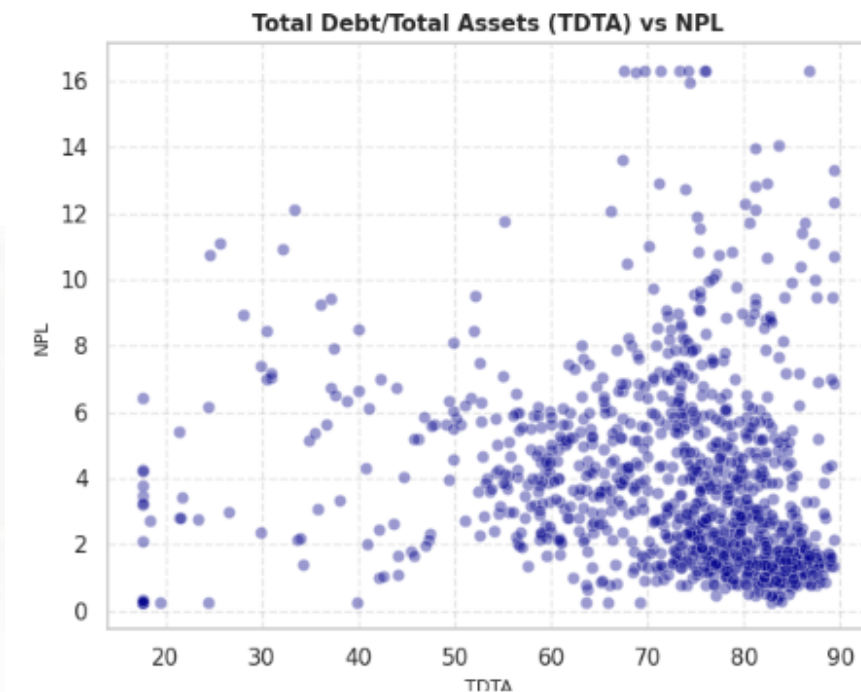
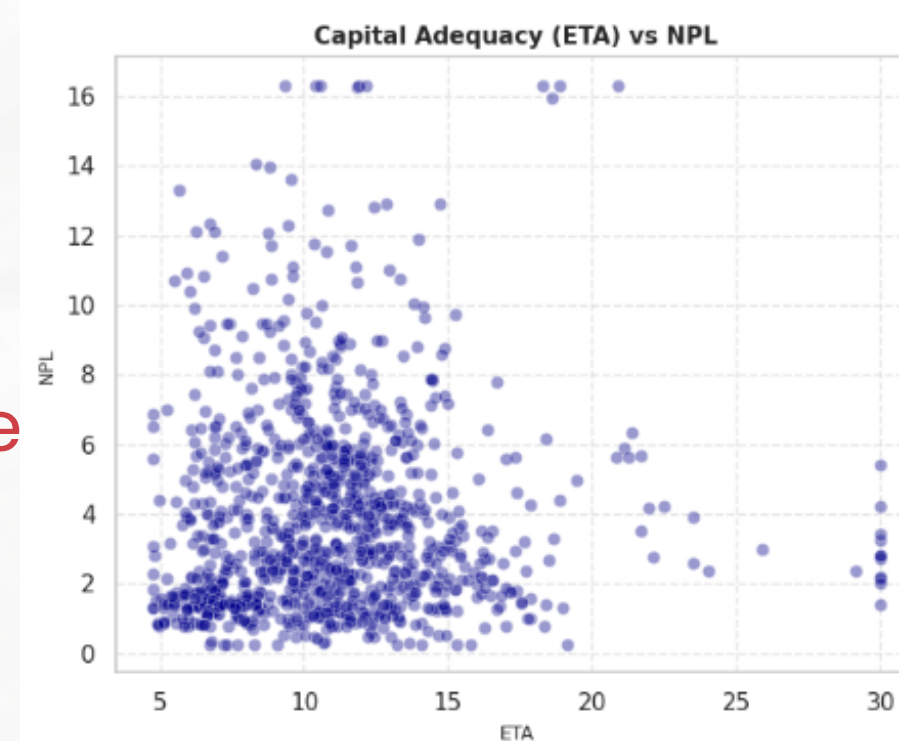
	Variable	VIF
0	const	617.927082
1	ESGS	1.288460
2	ETA	2.046556
3	CTI	1.300589
4	TDTA	2.009954
5	REVDIV	2.963079
6	LOANGR	1.152181
7	LNTA	1.563570
8	EG	1.197075
9	INF	1.540208
10	HHII	1.229795

Kết quả thống kê tính dừng ADF cho toàn bộ các ngân hàng:
Tổng số ngân hàng trong mẫu: 156
Số ngân hàng có chuỗi NPL KHÔNG DỪNG (p-value > 0.05): 139 (89.10%)
Số ngân hàng có chuỗi NPL DỪNG (p-value <= 0.05): 16 (10.26%)
Số ngân hàng không đủ điều kiện kiểm định (Dữ liệu không biến động/Quá ngắn): 1 (0.64%)



DESCRIPTIVE ANALYSIS & DATASET INSIGHTS

- NPL Asymmetry (Credit Risk) -> Highlights severe disparity and fragmentation in credit risk management across emerging institutions.
- High debt structure (TDTA mean = 72.16%) anchored by sound capital buffers (ETA mean = 10.79%).
- Wide variance in sustainability performance → Reflects highly polarized and unequal corporate sustainability integration.
- Micro factors display clear structural trends with NPL, whereas macro indicators appear highly diffused.
- Internal bank-level governance and operational attributes exert a more direct, critical footprint on asset quality than external macroeconomic shocks.



DESCRIPTIVE STATISTICS:

	Count	Mean	Std	Min	25%	50%	75%	Max
NPL	1092.0	3.844644	2.874599	0.250000	1.640000	3.020000	5.292500	16.330000
ESGS	1092.0	52.099026	17.870727	10.420000	40.182500	52.133170	64.605000	88.540000
ETA	1092.0	10.792398	3.875278	4.740973	7.948264	10.580167	12.607397	29.983417
CTI	1092.0	43.540186	12.697978	18.930000	33.440000	44.000000	51.642500	86.650000
TDTA	1092.0	72.156666	14.218892	17.574467	66.565225	75.896341	82.083485	89.355478
REVDIV	1092.0	10.207899	5.894785	3.981827	6.546274	8.780358	11.997473	44.125870
LOANGR	1092.0	10.674172	11.841454	-13.390000	3.777500	8.580000	15.062500	75.120000
LNTA	1092.0	24.690782	1.520446	21.521350	23.609398	24.638735	25.423372	29.027750
EG	1092.0	3.081432	3.942941	-9.518295	1.313914	3.297641	5.945208	11.439385
INF	1092.0	3.623866	4.005080	-2.093333	1.593136	2.811841	4.470367	19.873860
HHII	1092.0	12.024275	5.899953	5.371349	7.461799	9.912565	15.434650	28.213970



● S3: Designing the Panel Data Model

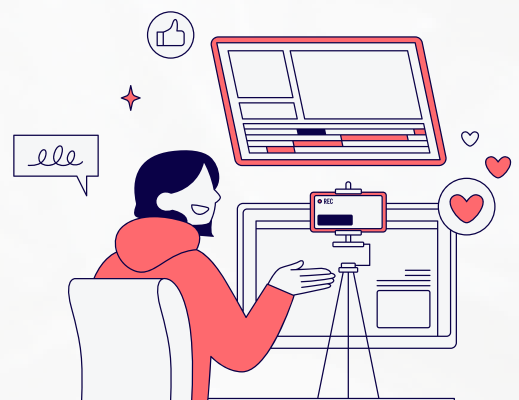
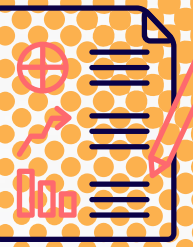
- Can ESG Mitigate Credit Risk? Evidence from Emerging Market Banks — A System GMM Approach
- Data: $156 \text{ banks} \times 7 \text{ years (2016–2022)} = 1,092$ observations

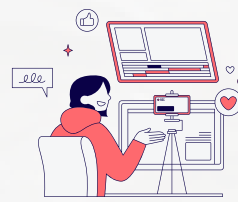
Static Model: Specification & 3 Tests

$$\text{NPL}_{it} = \beta_0 + \beta_1 \text{ESGS}_{it} + \beta_2 \text{ETA}_{it} + \beta_3 \text{CTI}_{it} + \beta_4 \text{TDTA}_{it} + \beta_5 \text{REVDIV}_{it} + \beta_6 \text{LOANGR}_{it} + \beta_7 \text{LNTA}_{it} \\ + \beta_8 \text{EG}_{it} + \beta_9 \text{INF}_{it} + \beta_{10} \text{HHI}_{it} + \mu_i + \varepsilon_{it}$$

TEST	COMPARISION	RESULT	CONCLUSION
F-test	Pool OLS vs FEM	$p < 2.2e-16$	Reject Pool OLS
Breusch-Pagan LM	Pool OLS vs REM	$p < 2.2e-16$	REM is significant
Hausman-test	FEM vs REM	$p = 1.38e-10$	Reject REM

FEM is the most appropriate static estimator according to the model-selection tests





Comparing 4 FEM Variants

Which estimation approach best fits this panel structure

LSDV



156 bank dummy variables added to OLS

Mathematically equivalent to Within, but computationally inefficient

BE



Regression on bank-level time averages

Captures only between-bank variation, ignores time dynamics, biased under endogeneity.

TWFE



Bank FE + Year FE simultaneously

*EG and INF are perfectly collinear with year dummies
→ VCOV not positive definite
→ macro coefficients wiped out.*

WE + HC3



Within Estimator – bank FE only, all 10 regressors retained

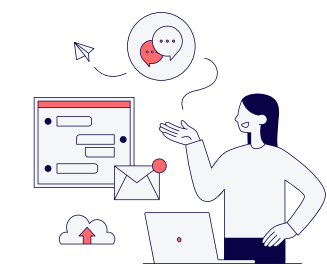
Selected

Preserves_macro variables (EG, INF; Robust HC3-SE_corrects heteroskedasticity/ autocorrelation)

VIF < 3 for all 10 regressors – multicollinearity in TWFE stems from year-dummy structure, not the underlying data



Final Static Model — Within-FEM with Robust HC3



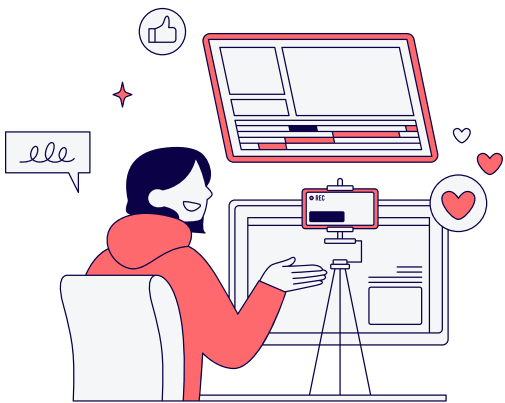
Variable	Coefficient	p-value	Interpretation
REVDIV	-0.1573 *	0.0312	Revenue diversification lowers NPL
LOANGR	-0.0266 ***	<0.0001	Loan growth dilutes NPL ratio (denominator effect)
EG	-0.0231 *	0.0277	Economic growth reduces default risk
ESGS	-0.0124 (n.s.)	0.1278	Negative sign as expected, not significant here



Robustness check (TWFE, EG/INF excluded, clustered SE): ESGS = -0.0172* , LOANGR = -0.0261*** , REVDIV = -0.1681*

Core coefficients retain sign and significance across specifications, confirming robustness. ESGS remains negative across specifications, although statistically insignificant

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.05$, n.s. = not significant



Dynamic Panel Model — Why System GMM?

Corr(NPL, NPL_lag1)

0.9184

Evidence of strong persistence in NPL

Two-step System GMM

transformation = "ld"

Combines level and difference equations; instruments via lag(var, 2:4); controls bank FE + year FE.

NPL exhibits strong persistence — past NPL drives current NPL (consistent with "bad management" / "evergreening" theory). A static model omits this dynamic, causing omitted variable bias.

$$NPL_{it} = \alpha \cdot NPL_{it-1} + \beta_1 ESGS_{it} + \Sigma Controls_{it} + \mu_i + \varepsilon_{it}$$

System GMM Results & Diagnostics

lag(NPL, 1) coefficient

0.9266 ***

Speed of adjustment

≈ 7.34% / year

Test	Result	Status
Sargan test	p = 0.139	Valid 
AR(1)	p = 0.002	OK 
AR(2)	p = 0.015	Limitation 



Limitation: AR(2) remains significant at the 5% level (p = 0.015), suggesting the lag(2:4) instrument set does not fully eliminate second-order autocorrelation.

To be addressed in S4 by shifting the instrument lag range to lag(3:5) or lag(4:6).

Sargan test confirms instrument validity; AR(1) presence is expected and not a concern.

S3 Process Summary

STATIC TRACK

EDA

3 selection tests

Compare FEM variants

Within-FEM + HC3

DYNAMIC TRACK

High lag correlation (0.92)

System GMM (2-step)

Sargan / AR(1) / AR(2)

Speed of adjustment 7.34%

Combining static and dynamic approaches provides a more comprehensive view of the relationship between ESG performance and credit risk.

EMPIRICAL RESEARCH ASSESSMENT

Econometric Model Diagnostics & Error Handling

Empirical Modeling Assessment: Static Panel Frameworks vs. Dynamic System GMM

STATIC PANEL DIAGNOSTICS

1. Estimator Selection Flow

Rigorous comparative testing identifies the Fixed Effects Model (FEM) as the most consistent static estimator.

- **F-test & BP-LM Tests:** Both diagnostic tests yield extreme significance levels $p < 2.2 \times 10^{-16}$ decisively rejecting Pooled OLS in favor of unobserved individual effects.
- **Hausman Test:** Achieves absolute statistical significance with $p = 1.379 \times 10^{-10}$ rejecting Random Effects (REM) due to systematic endogeneity.
- **Individual Heterogeneity:** Permanent structural differences across the 156 banks over 7 years require unobserved bank-level fixed effects.

2. Diagnostics & Remedies

Violations of basic Gauss-Markov static model assumptions were identified and systematically resolved:

- **Multicollinearity:** Variance Inflation Factors (VIF) are strictly less than 3 for all independent variables, confirming no critical collinearity concerns.
- **Heteroskedasticity & Autocorrelation:** Non-constant variance and error correlation were successfully addressed by implementing Robust HC3 standard errors via R's `vcovHC()`.
- **TWFE Macro-Collinearity Trap:** Adding Year Fixed Effects creates perfect collinearity with macroeconomic variables (EG, INF) changing only by year, yielding a Singular Matrix error.
- **Remedy Action:** Prioritize the 1-Way Within-FE model combined with Robust HC3 as the baseline; reserve TWFE exclusively for checking robustness.

SYSTEM GMM DIAGNOSTICS

0.9184

NPL First-Lag Correlation

The extreme persistent inertia of non-performing loans mandates a Dynamic Panel GMM framework to effectively control for Nickell Bias present in static FEM models.

DIAGNOSTIC TEST	STATISTIC/P-VALUE	STATUS	ECONOMETRIC INTERPRETATION
Arellano-Bond AR (1)	p = 0.00161	PASSED	Confirms first-order autocorrelation in first differences. This is a vital necessary condition for proper GMM specification.
Sargan Test	p = 0.13849	VALID	Fails to reject H0; over-identifying instrumental variables satisfy exogeneity requirements and are fully valid.
Arellano-Bond AR (2)	p = 0.01458	ERROR	Detects second-order autocorrelation in residuals. This systematic error invalidates second-order lags as instruments.

Model Optimizations

Maintain 1-Way Within-FE

1

Utilize the 1-way Fixed Effects model with Robust HC3 corrections as the core static baseline. This avoids singular matrix errors, successfully preserving macroeconomic indicators (EG, INF) for structural analysis.

Recalibrate Instrument Lags

2

To eliminate the systematic $AR(2)$ serial correlation, adjust the instrument lag window in the R `pgmm` command. Shift the lag range from the current default 2:4 to either 3:5 or 4:6 to restore instrument validity

Prioritize Two-step GMM

3

Two-Step System GMM remains the theoretically optimal estimator for final policy inferences due to strong NPL inertia and endogeneity risks. It is mathematically robust once the lag adjustments from Step 2 are implemented.



Determinations & conclusion

Determinants of Credit Risk

A Dynamic Panel Data Analysis of Non-Performing Loans (NPL) with ESG and Revenue Diversification Factors

MODEL SELECTION & EVIDENCE

Why System GMM?

Static models (Pooled OLS, FEM) fail to capture the **dynamic nature** of banking risks and suffer from endogeneity.

Statistical Test	Result / R Evidence
F-test	$F = 20.957(p < 2.2e - 16) \rightarrow \text{Reject Pooled OLS}$
Hausman Test	$\chi^2 = 67.441(p < 0.001) \rightarrow \text{Reject REM}$
Correlation	$\text{Corr}(NPL, NPL_{lag1}) = 0.9184$
Persistence	$NPL_{lag1} = 0.9266$



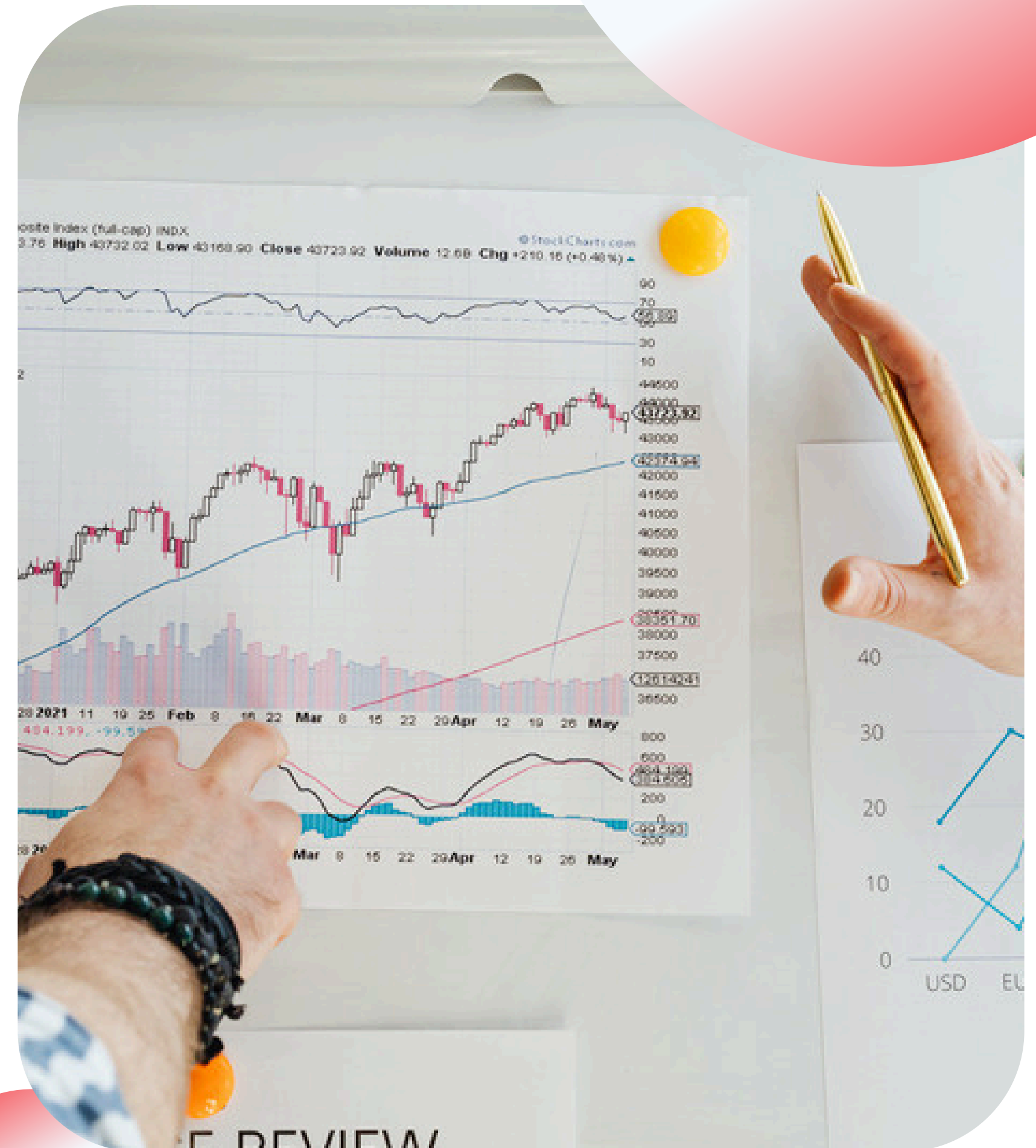
POWER OF PERSISTENCE



0.9266

CARRY-OVER COEFFICIENT

Economic Reality: Bad loans are not isolated shocks. Approximately **92.66%** of today's NPL carries over into the next period. Accumulation causes high persistence through debt restructuring costs, collateral liquidation delays, and reduced lending capacity.



THE DIVERSIFICATION PARADOX

The sign reversal between **Fixed Effects (FEM)** and **System GMM** reveals the critical importance of endogeneity.

FEM Perspective (Short-term)

-0.1573*

Suggests diversification is **beneficial**. More non-interest income reduces dependence on lending and stabilizes earnings.

GMM Perspective (Long-term)

+0.0666*

Reveals diversification **increases risk** over time. Operational complexity and unfamiliar activities create governance challenges.

Finding: Diversification without risk-governance creates new risks rather than reducing existing ones.

ESG & INSTITUTIONAL RESILIENCE



Long-term Factor Impact

While ESG loses significance in the dynamic GMM model ($p=0.317$), the coefficient remains **consistently negative (-0.0056)**.

- 🛡️ **Indirect Influence:** ESG strengthens risk culture and transparency rather than causing immediate NPL reductions.
- 🏛️ **Institutional Factor:** It acts as a long-term buffer for bank resilience during macroeconomic shocks.

STRATEGIC POLICY OUTLOOK

Prevention Over Resolution

The persistence effect (0.9266) dominates all other determinants (ESG, REVDIV, EG). Stability depends on **path-dependency**.

"Preventing NPL accumulation is substantially cheaper and more effective than resolution mechanisms."

- **Early Warning Systems:** Proactive monitoring.
- **Forward Provisioning:** Anticipating the cycle.
- **Timely Restructuring:** Preventing defaults.



MULTIPLE CHOICE QUESTION



R Output 5 - System GMM Diagnostic Tests

System GMM diagnostic tests

Sargan test: $\text{chisq}(116) = 132.6336$ (p-value = 0.13849)

Autocorrelation test (1): normal = -3.154168 (p-value = 0.0016096)

Autocorrelation test (2): normal = -2.442612 (p-value = 0.014581)

Wald test for coefficients: $\text{chisq}(11) = 3512.798$ (p-value = $< 2.22\text{e-}16$)

Wald test for time dummies: $\text{chisq}(5) = 8.314061$ (p-value = 0.13976)

Question 1. Based on the System GMM diagnostic tests shown in the R output, consider the statements below:

- I. The Sargan test does not reject the validity of instruments at the 5% level.
- II. A significant AR(1) test is usually acceptable in Difference/System GMM.
- III. The AR(2) test is ideal because its p-value is smaller than 0.05.
- IV. The significant AR(2) result suggests that the GMM specification may need further refinement.

A

I, II, and IV only

B

I and III only

C

II and III only

D

I, II, III, and IV

Solution: I, II, and IV only

EXPLANATION

Statements I, II, and IV are correct. The Sargan test p-value is 0.13849, so instrument validity is not rejected at 5%. Significant AR(1) is common after differencing. However, AR(2) should ideally be insignificant. Since AR(2) p-value = 0.014581, this suggests possible second-order serial correlation and the specification may need refinement

R Output 2 - FEM Robust and TWFE Coefficients				
FEM with robust standard errors (HC3)				
	Estimate	Std. Error	t value	Pr(> t)
ESGS	-0.0124062	0.0081395	-1.5242	0.12780
REVDIV	-0.1573415	0.0729353	-2.1573	0.03124 *
LOANGR	-0.0266499	0.0048177	-5.5317	4.127e-08 ***
EG	-0.0230677	0.0104604	-2.2052	0.02768 *
INF	-0.0470601	0.0255767	-1.8400	0.06609 .
Two-way Fixed Effects (Bank_ID + Year), clustered by Bank_ID				
Observations: 1,092 Fixed-effects: Bank_ID: 156, Year: 7				
	Estimate	Std. Error	t value	Pr(> t)
ESGS	-0.017157	0.008598	-1.9954	0.047752 *
ETA	0.109750	0.053226	2.0620	0.040882 *
REVDIV	-0.168140	0.072437	-2.3212	0.021579 *
LOANGR	-0.026135	0.004699	-5.5623	1.1405e-07 ***
RMSE: 1.20486 Adj. R2: 0.79193 Within R2: 0.076984				

Question 2. Based on the FEM with robust standard errors (HC3) result in the R output, consider the statements below:

- I. ESGS has a negative coefficient but is not statistically significant at the 5% level.
- II. REVDIV has a negative and statistically significant coefficient at the 5% level.
- III. LOANGR has a negative and highly significant coefficient.
- IV. INF is statistically significant at the 1% level.

A

1

B

2

C

3

D

4

Solution: 3

EXPLANATION

Statements I, II, and III are correct. ESGS is negative but $p\text{-value} = 0.12780$, so it is not significant at 5%. REVDIV is significant at 5%, and LOANGR is highly significant. INF has $p\text{-value} = 0.06609$, so it is only weakly significant at the 10% level, not at the 1% level.

Question 3. Given:

FEM: REVDIV=-0.1573

System GMM: REVDIV=+0.0666

Evaluate:

- (1) The relationship between diversification and credit risk depends on model specification.
- (2) Diversification may generate both risk-reducing and risk-increasing mechanisms.
- (3) The sign reversal highlights the importance of considering dynamic persistence and endogeneity.
- (4) The sign reversal proves that one of the two models is invalid.

How many statements are correct?

A

1

B

2

C

3

D

4

Solution: 3

EXPLANATION

Statements I, II, and III are correct. The trap is subtle. Must understand why sign changes occur in empirical finance.

Thank You

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```

> model_sgmm_fixed2 <- pgmm(NPL ~ lag(NPL, 1) + ESGS + ETA + CTI + TDTA + REVDIV + LOANGR + LNTA + EG + INF + HHII |
+                               # Chỉ giữ lại các biến thực sự nội sinh mạnh) và lùi trễ 3:4
+                               lag(NPL, 3:4) +
+                               lag(LOANGR, 3:4) +
+                               lag(REVDIV, 3:4) |
+                               # Đưa các biến kiểm soát khác và biến vĩ mô làm ngoại sinh sau dấu | thứ hai
+                               ESGS + ETA + CTI + TDTA + LNTA + EG + INF + HHII,
+ data = pdata,
+ effect = "twoways",
+ model = "twosteps",
+ transformation = "ld")
> print(summary(model_sgmm_fixed2, robust = TRUE))

```

Twoways effects Two-steps model System GMM

Call:

```

pgmm(formula = NPL ~ lag(NPL, 1) + ESGS + ETA + CTI + TDTA +
      REVDIV + LOANGR + LNTA + EG + INF + HHII | lag(NPL, 3:4) +
      lag(LOANGR, 3:4) + lag(REVDIV, 3:4) | ESGS + ETA + CTI +
      TDTA + LNTA + EG + INF + HHII, data = pdata, effect = "twoways",
      model = "twosteps", transformation = "ld")

```

Balanced Panel: n = 156, T = 7, N = 1092

Number of Observations Used: 1404

Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-10.184653	-0.468359	-0.016354	-0.009985	0.440254	11.347810

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)	
lag(NPL, 1)	0.9442092	0.0581879	16.2269	< 2.2e-16	***
ESGS	-0.0053436	0.0029333	-1.8217	0.0685013	.
ETA	-0.0325704	0.0272806	-1.1939	0.2325157	
CTI	-0.0030473	0.0048009	-0.6347	0.5256039	
TDTA	0.0143284	0.0091747	1.5617	0.1183531	
REVDIV	0.0688322	0.0396857	1.7344	0.0828413	.
LOANGR	-0.0078862	0.0099020	-0.7964	0.4257832	
LNTA	-0.0261725	0.0342002	-0.7653	0.4441084	
EG	-0.0470978	0.0136410	-3.4527	0.0005551	***
INF	-0.0214352	0.0174650	-1.2273	0.2197001	
HHII	0.0119147	0.0079848	1.4922	0.1356547	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sargan test: chisq(41) = 56.45567 (p-value = 0.054635)

Autocorrelation test (1): normal = -2.510678 (p-value = 0.01205)

Autocorrelation test (2): normal = -1.559068 (p-value = 0.11898)

Wald test for coefficients: chisq(11) = 1347.942 (p-value = < 2.22e-16)

Wald test for time dummies: chisq(4) = 3.767881 (p-value = 0.43833)

```

> model_sgm_fixed3 <- pgmm(NPL ~ lag(NPL, 1) + lag(NPL, 2) + ESGS + ETA + CTI + TDTA + REVDIV +
+ lag(NPL, 3:5) +
+ lag(ESGS, 3:5) +
+ lag(ETA, 3:5) +
+ lag(CTI, 3:5) +
+ lag(TDTA, 3:5) +
+ lag(REVDIV, 3:5) +
+ lag(LOANGR, 3:5),
+ data = pdata,
+ effect = "twoways",
+ model = "twosteps",
+ transformation = "ld"
+ )
> print(summary(model_sgm_fixed3, robust = TRUE))

```

Twoways effects Two-steps model System GMM

Call:

```

pgmm(formula = NPL ~ lag(NPL, 1) + lag(NPL, 2) + ESGS + ETA +
      CTI + TDTA + REVDIV + LOANGR + LNTA + EG + INF + HHII | lag(NPL,
      3:5) + lag(ESGS, 3:5) + lag(ETA, 3:5) + lag(CTI, 3:5) + lag(TDTA,
      3:5) + lag(REVDIV, 3:5) + lag(LOANGR, 3:5), data = pdata,
      effect = "twoways", model = "twosteps", transformation = "ld")

```

Balanced Panel: n = 156, T = 7, N = 1092

Number of Observations Used: 1404

Residuals:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	-10.61976	-0.51343	-0.04170	0.01359	0.47049	11.50902

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)	
lag(NPL, 1)	0.9923615	0.0657461	15.0939	< 2.2e-16	***
lag(NPL, 2)	-0.1904674	0.0561101	-3.3945	0.0006875	***
ESGS	-0.0074187	0.0063713	-1.1644	0.2442664	
ETA	-0.0039574	0.0419833	-0.0943	0.9249012	
CTI	0.0024438	0.0090230	0.2708	0.7865154	
TDTA	0.0136315	0.0060733	2.2445	0.0248002	*
REVDIV	0.0705936	0.0295036	2.3927	0.0167244	*
LOANGR	-0.0083211	0.0071629	-1.1617	0.2453563	
LNTA	-0.0536846	0.0504960	-1.0631	0.2877166	
EG	-0.0428242	0.0123971	-3.4544	0.0005516	***
INF	-0.0030800	0.0169510	-0.1817	0.8558197	
HHII	0.0098794	0.0132237	0.7471	0.4550047	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sargan test: chisq(94) = 103.3371 (p-value = 0.2395)

Autocorrelation test (1): normal = -3.900406 (p-value = 9.6032e-05)

Autocorrelation test (2): normal = -0.443926 (p-value = 0.6571)

Wald test for coefficients: chisq(12) = 1898.759 (p-value = < 2.22e-16)

Wald test for time dummies: chisq(4) = 9.219075 (p-value = 0.055851)